



Name: Santanu Sengupta

Contact: 341 777 6695

Email: santanusengupta876@gmail.com

LinkedIn: linkedin.com/in/santanu-sengupta-9870932

Lead java full stack developer

SUMMARY

- Experienced in over 15years designing and implementing real-time data processing pipelines using Apache Spark Framework on Azure/AWS Databricks.
- Proficient in reactive programming frameworks for building responsive and scalable systems.
- Expertise in Azure cloud services, including Blob Storage, Service Bus, Key Vault, and Azure Active Directory.
- Skilled in leveraging AWS services such as S3, EC2, and Lambda to build scalable, cost-efficient backend infrastructure.
- Recognized for delivering innovative solutions to streamline operations and enhance system performance.
- Used Bootstrap and AngularJS, React.js and Node.js in effective web design.
- Developed Kafka consumers to consume data from Kafka topics. Used Bootstrap and AngularJS, React.js and Node.js in effective web design.
- Led a team to develop a robust web application using Angular and React, achieving a 20% improvement in user engagement.
- Possess a visionary mindset with the ability to quickly grasp business details and deep technical understanding.
- Experience in Spring Framework such as Spring IOC, Spring DAO, Spring ORM, Spring Resources, Spring JDBC, Spring Boot and Micro - Services.
- Experienced in developing standalone applications using Spring Boot.
- Experience in Spring Framework modules such as Spring MVC, IOC, AOP, JDBC, JTA, IO, Spring Dashboard, Spring Boot and Spring Micro services, Spring REST
- Experienced in developing web applications with Single Page Applications (SPAs) Framework using Angular.JS.
- Experienced in working with React.JS and Redux
- Sound knowledge of architectural patterns, best practices, and non-functional requirements.
- Skilled in developing monolithic applications and microservices architectures using REST APIs, Spring Boot, Reactive Frameworks, and Java 17.
- Designed and optimized real-time structured streaming and batch ETL pipelines using Java, Scala, Python, and Delta Lake on Databricks.
- Expertise in database management, including NoSQL (Cassandra, Cosmos DB, MongoDB, DynamoDB) and relational databases (RDS, SQL Server, MySQL, DB2).
- Proficient in DevOps practices, including CI/CD automation using Jenkins, Kubernetes, Pivotal Cloud Foundry, Fargate, and GitHub Actions.
- Conducted root cause analysis to identify and resolve performance bottlenecks, memory leaks, and system inefficiencies.
- Guided and mentored junior engineers in Azure Databricks best practices, client-specific use cases, and efficient data processing.
- Collaborative team player, skilled in prioritizing platform features, analyzing cost-benefit solutions, and effectively communicating across cross-functional teams.

TECHNICAL SKILLS

Operating Systems	Windows, Linux
Languages	Java 8/11/17
Frameworks	Spring, Spring MVC, SpringBoot, Struts, JSF, JPA/Hibernate, iBATIS, Spring Security, Microservices, Spring Cloud

Web Technologies	HTML, JavaScript, CSS, JSON, XML, Ajax, jQuery, Angular v6/7/8/9/10, Angular JS, React, Node.js
Server Scripting	JSP, JSP-EL, JSTL, Custom Tags
Client Scripting	JavaScript, jQuery, Angular, React
Web Services	SOAP, UDDI, WSDL, JAX-RS, JAX-WS, REST API
Application/web Servers	WebLogic, IBM Web Sphere, JBoss, Jetty, Tomcat
Message Queue Servers	IBM Message Queue, Kafka, Active MQ, RabbitMQ
Databases	Oracle, MS SQL Server, PostgreSQL, Mongo DB, Cassandra
Database Tools	PL/SQL Developer, PostgreSQL Client, DBeaver, SSMS, Mongo DB compass, Studio 3T
Data Query Language Technologies	SQL, NoSQL, GraphQL, Elasticsearch Query DSL, Amazon DynamoDB
Cloud Computing	AWS - EC2, Lambda, S3, Elastic Beanstalk, RDS, CloudFormation, Elastic Load Balancing, IAM, SQS, CloudWatch, API Gateway, Azure, Pivotal Cloud Foundry
Data Steaming Technologies	Apache Kafka, Apache Flink
Containerization Tools	Kubernetes, Docker
Continuous Integration	Jenkins, Hudson, Bamboo
Observability and Monitoring Tools	Splunk, Prometheus, and Grafana
Authentication and Authorization Technologies	Basic Authentication, Token-Based Authentication (JWT), OAuth, LDAP, MFA.
Version Control	VSS, TFS, Tortoise SVN, Git, GitLab, and Azure Repos
Development Tools (IDEs)	Eclipse, My Eclipse, Visual Studio code, IntelliJ IDEA, STS
Build Tools	Ant, Maven, Gradle, Log4j, SonarQube, Junit, Protractor, Mockito, SOAP UI, Postman, YAML
Code Review Tools	Atlassian Crucible
Documentation Tools	Swagger, Confluence
Testing Tools	Junit, Mockito, Jacob Co, JMeter, and Selenium
Project Management Tools	Agile (XP, Scrum), Rational Unified Process (RUP), Waterfall

Education

- B.E in Information Technology, India.

Certifications

- Certified with Databricks Certified Associate Developer for Apache Spark 3.0 in Scala
- AWS Certified Cloud Practitioner
- Certified (WebSphere Commerce Application Developer v.6 & v.7) in WebSphere Commerce from IBM

PROFESSIONAL EXPERIENCE

Albertsons, Pleasanton, California
Technical Lead

Sep 2022 - Present

Responsibilities:

- Enterprise Gamification for Associates: Designed and developed an enterprise gamification platform to boost engagement among associates and customers. Built an admin interface for game management, integrated multiple data sources, and implemented leaderboards. Utilized a modular system architecture with plug-in support for diverse use cases. Developed real-time streaming and batch processing workflows in Databricks, ensuring system flexibility and scalability using Java, Spring Boot, Scala, Python, Spark, and Atlas MongoDB.
- A/B Testing and Experimentation: Architected and implemented an in-house A/B testing solution that enhanced user experiences, driving a 25% increase in customer engagement. Streamlined product decision-making processes by 40% through efficient API integration with the company's ecosystem, achieving improved efficiency and cost savings.
- Using Node JS for server-side code and written callback functions for the Rest Full API.
- Expertise in using Node JS, Express JS, and Mongo DB along with Mongoose.JS to create server-side applications
- Build prototype for various required services such as Scheduling, Logging and Notification Service using third party Node JS based JavaScript library
- Built main application leveraging technologies such as Angular.js and used NodeJS to run angular.
- Implemented logger for various core services using Spring AOP and log4j.
- Built the internal tool and maintained it with React.JS along with Flux and Redux.
- Developed UI using React.js and Redux and Next.js.
- Implemented modules into Node.JS to integrate with the design and requirements.
- Designed and developed Micro-services using REST framework and Spring Boot and used Bootstrap along with React.JS in creating the Application.
- Developed Java 17 applications, leveraging modern features and enhancements.
- Developed responsive JEE Web Applications using Java 17 and cutting-edge technologies
- Designed and developed the REST based Micro services using the Spring Boot, Spring Data with JPA
- Omni-Channel AcuStock Inventory Management: Led the design and deployment of an inventory management system for Wine Fulfillment Centers (WFCs), enhancing operational workflows. Enabled store associates to perform tasks like Put Away, Inventory Adjustments, Cycle Counting, and barcode scanning for item details. Delivered a dedicated inventory management system aligned with MLP requirements, allowing associates to submit inventory snapshot requests seamlessly.
- Enterprise Promotion Engine (Loyalty): Designed and deployed multiple Spark-based real-time streaming jobs, boosting data processing capacity and handling over 2 million data points daily. Enhanced analytics operations and system performance through efficient real-time and batch processes for offer redemption and continuity offers.

Environment: Java/Spring Boot/Scala/Python | Apache Spark | DataBricks Batch/Structured Streaming | Azure, AWS Cloud | Microservices | Reactive Framework | WebFlux | Rest API | Distributed Programming | Big Data | KAFKA

Lows (IBM), Mooresville, North Carolina
Technical Lead

Aug 2016 - Sep 2022

Responsibilities:

- Enterprise Promotion Engine (Offer Generation): Modernized application architectures using reactive programming, microservices, and secured APIs, enhancing scalability and efficiency. Built a comprehensive offer management platform to manage promotions, enabling functionalities such as offer allocation, reporting, personalization, pricing maintenance, and support team access for issue resolution.
- Customer & Communications: Integrated OKTA for identity management, implementing OAuth2 and OpenID Connect frameworks for secure data access. Strengthened system security by implementing role-based access control (RBAC) and multi-factor authentication (MFA). Improved user experiences with single sign-on (SSO) capabilities and ensured compliance with industry standards through collaboration with security teams.
- Working with Bootstrap Components which are written in AngularJS using Angular-UI components
- Used various features in AngularJS like data binding, routing, custom filters, custom directives to meet various customized requirements.
- Provided client side validations using AngularJS framework
- Worked with AngularJS for creating their own HTML elements for building a company's site which is an open Source Project.
- Led the WCS development team for Lowe's, managing the development and launch of a B2B website integrated with Lowe's infrastructure, and provided technical leadership to offshore teams.
- Involved in various phases of Software Development Life Cycle, such as requirements gathering, modeling, analysis, design and development.

- Developed application using Java 17 to implement its features like lambdas Expressions, Time API, Streams, functional interfaces, collectors, default methods, type interfaces, for each.
- Developed the application using Spring Framework Model View Controller (MVC) architecture.
- Worked with controller, service and view components in applications using spring framework.
- Developed the business layer using Spring Boot.
- Used Express Framework to set up middleware to respond to the HTTP requests with the help of Node JS.
- Developed web page using React JS, Redux and Node JS
- Used Mongo DB along with Node JS, as a database to store JSON.
- Used Spring Boot framework to create properties for various environments and for configuration.
- Implementing or exposing the Micro services to base on RESTful API utilizing Spring Boot.
- Developing new user-facing features using React.JS.
- Developed a UI Component Using React.JS and Redux.
- Developed front-end user interfaces with React.JS by creating many components and modules.
- Used React.JS to create views to hook up models to the DOM and synchronize data with server as a Single Page Application (SPA).
- Developed REST services for critical modules, created JUnit test cases ensuring Sonar code compliance, and implemented performance-tuning strategies to optimize system efficiency.
- Managed bi-weekly development releases adhering to agile methodology, collaborated with multiple teams across product lifecycles, and streamlined development processes for efficient product deployments.
- Coordinated with business teams to gather requirements, ensured code quality through rigorous testing, and delivered status reports and updates to project stakeholders and senior leadership.
- Containerized Micro services by creating Docker images from Docker file collaborated with development support team to set up a continuous deployment environment using Docker.
- Used EJB as a middleware in designing and developing a three-tier distributed application.
- Used Log4j as a debugging tool for logging application debugging messages.
- Implemented test cases and performed automation testing using JUnit Framework
- Continuous Integration (CI) and Continuous Delivery (CD) process implementation using Jenkins.
- Developed both Restful and SOAP web services depending on the design need of the project.
- Worked with MongoDB to store non-relational data into collection and retrieve them whenever required.
- Written SQL Queries and PL/SQL stored procedures to satisfy various business requirements of the application.
- Used Agile methodology for developing the application.
- Actively participated and provided feedback in a constructive and insightful manner during weekly Iterative review meetings to track the progress for each iterative cycle and figure out the issues.

Environment: Java/Spring Boot | Amazon DynamoDB, RDS, RedShift | SQL Server | KAFKA | Node.js | Full Stack | System Design Thinking

Staples (IBM), Framingham, Massachusetts
Senior java full stack developer

Feb 2015 – Jul 2016

Responsibilities:

- Led development efforts for e-commerce solutions using WebSphere Commerce Suite, optimizing core Java/J2EE applications and enhancing performance and scalability.
- Conducted requirement gathering, Fit-Gap analysis, and collaborated with clients and stakeholders to define project requirements and deliverables, ensuring alignment with client needs.
- Led teams in building complex enterprise applications, ensuring high code quality, system reliability, and seamless integration with existing enterprise systems.
- Used SVN and Source Tree for configuration management, performed performance tuning and testing, and delivered regular updates to senior management on project progress and performance metrics.
- Used React.JS for templating for faster compilation and developing reusable components and utilized react for its efficient data flow architecture to create a lightweight application.
- Developed React.JS applications using Redux architecture for front-end UI, and Node.js applications for backend.

- Used Node.JS as a proxy to interact with Rest services and interacting with Database.
- The Java Message Service (JMS) API used to allow application components to create, send, receive, and read messages.
- Enhance the applications with new add - on features, Java 14, TypeScript 2.8, .Net, React.JS, Jenkins using Angular9 on front-end and Node JS on back-end with ECMAScript 6(ES6) features.
- Worked throughout the full stack (Node JS, Angular JS) to see software extensions through to completion.
- Developing Docker images for Development and Testing teams and their pipelines.
- Used AWS to deploy the project on an EC2 instance. Created highly fault tolerant, highly scalable Java application using AWS Elastic Load Balancing, EC2, VPC and S3 as part of process and security improvements.
- Used Log4j to capture the log that includes runtime exceptions and debug information.
- Involved in developing JUNIT Test Cases to validate the type of data in the XML Files.
- Used Jenkins for building, creating snapshots and deploying on servers.
- Implemented Kafka Streams for real-time processing and transformation of streaming data using Java.
- Built CI/CD pipeline and configured all the credentials and users in Jenkins.
- Responsible for orchestrating CI/CD processes by responding to GIT triggers, human input, and dependency chains and environment setup.
- Made use of Rx.js observables in Angular 4 for setting up a direct pipe in-order to request data from the back-end.
- Optimized Neo4j database performance through indexing, query optimization, and cache management to ensure scalability and fast data processing.
- Implemented Service Oriented Architecture (SOA architecture) using Web Services (SOAP, REST).
- Worked on MongoDB design and indexing techniques. Implemented read preferences in MongoDB replica set.
- Involved in Stored Procedures, User Defined functions, Views and implemented the Error Handling in the Stored Procedures and SQL objects and modified already existing stored procedures, triggers, views, indexes depending on the requirement.
- Followed AGILE Methodology to promote iterations, collaboration, and process adaptability and participated in SCRUM Meetings throughout the life cycle of project

Environment: Java | Websphere Commerce Server | DB2

Micron, Husqvarna, Whirlpool Corporation (IBM), Kolkata Area, India
Senior java developer

Jun 2010 – Feb 2015

Responsibilities:

- Led development teams in delivering scalable e-commerce solutions using WebSphere Commerce Suite for clients like Micron and Whirlpool, ensuring project success through requirement gathering, Fit-Gap analysis, and managing code and configuration changes.
- Developed and distributed real-time data feeds for catalog and pricing subsystems at Husqvarna and managed a 10–15-person team for B2B and B2E web warranty integration projects.
- Designed and implemented real-time data processing systems, enhancing operational efficiency, and provided day-to-day directions to development teams to maintain project timelines and quality.
- Collaborated with stakeholders to design and deliver high-performance solutions, ensuring smooth integration of data feeds and business logic, and managed scope, planning, and tracking for multiple projects.

Environment: Java | Websphere Commerce Server | DB2

Key Management Group Inc
Java developer

Jun 2006 - Jun 2010

Responsibilities:

- Led the development of new modules and enhancements. Conducted thorough unit testing, identifying and resolving issues to ensure the reliability and stability of the software. Maintained consistent formatting and style throughout the codebase, enhancing readability and maintainability. Actively participated in code reviews, providing valuable feedback and suggestions for improvement to team members.

Environment: Java | DB2